TeDFL: Toward Text-Driven Few-Shot Learning for Forest Tree Species Classification With Airborne Hyperspectral Images

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***Abstract*—Tree species classification is fundamental for forest inventory and sustainable silviculture. Airborne hyperspectral images (HSIs) can capture subtle spectral differences between tree species through dense spectral channels, aiding the discrimination of similar species. However, in multispecies field scenarios, mixed pixels and environmental interference exacerbate interclass spectral similarity and intraclass spectral variation, making it difficult to accurately distinguish tree species solely through visual representation. To this end, a text-driven few-shot learning (TeDFL) paradigm is developed that can leverage linguistic prior knowledge to guide the generation of discriminative visual features. The TeDFL method aims to train a highly flexible and robust model that can adapt to classifying diverse, yet similar tree species using just a few examples. Our TeDFL framework contains two primary encoders: a visual encoder and a text encoder. The former exploits a ghost attention network (GANet) to explore spatial–spectral visual features without excessive parameters, where the GANet consists of ghost- and normalization-based attention blocks. The latter leverages the contrastive language–image pretraining (CLIP) model to incorporate additional rich semantic priors, enriching the visual feature space. Furthermore, we design cross-modal alignment (CMA) and cross-modal integration (CMI) strategies to bridge the modal discrepancy between visual and textual representations. The CMA strategy aims to facilitate a visual–text contrastive objective by contrasting HSIs with class-specific text prompts, while the CMI strategy adaptively fuses multimodal support data to obtain more reliable prototypes. Experiments on four realistic forest scenarios confirm the feasibility and effectiveness of the TeDFL method compared with other state-of-the-art methods. Moreover, the TeDFL method could serve as a standard visual–language backbone network for few-shot tree species classification research. The source code will be made publicly available at https://github.com/HlEvag/TeDFL.git**

***Index Terms*—Airborne hyperspectral images (HSIs), few-shot learning (FSL), linguistic prior knowledge, multimodal, tree species classification.**

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I. Introduction

FOREST tree species classification is a fundamental task in forestry, which aims to identify and classify tree species within forest ecosystems. It plays a significant role in various aspects of forest management, environmental monitoring, and biodiversity conservation [1]. However, similarities in morphology, color, and phenology among tree species hinder accurate identification, thereby increasing measurement errors and resource costs for forest inventory. Airborne hyperspectral images (HSI) can offer nanoscale spectral resolution, which can finely depict the subtle differences among various land objects [2]. These images can provide detailed insights into the composition and distribution of tree species at the pixel level, offering vital evidence for ecological management and policy-making.

High-dimensional airborne hyperspectral data acquired from forest scenes often involve massive data volumes and high computational complexity [3], [4], [5]. Constrained by this dilemma, early studies typically performed feature extraction before pixel classification to filter out more significant features from the raw cube data. Common feature extraction approaches are separated into linear operations, such as principal component analysis [6] and linear discriminant analysis [7], and nonlinear operations like local linear embedding [8] and manifold learning [9]. Through these methods, we can input key features of various tree species into k-nearest neighbor [10], extreme learning machine [11], or support vector machine [12] classifiers to obtain prediction results for each pixel. However, these conventional techniques struggle to handle diverse tree species because they rely on manual spectral feature design. Over the last decade, convolutional neural networks (CNNs) have gradually become a mainstream deep learning (DL) tool due to their flexibility and adaptability [13], [14]. Hamida et al. [15] adopted a 3-D deep learning (3DDL) approach to harness the advantages of spatial–spectral features. Zhang et al. [16] developed a modified 3D–1D-CNN model for tree species classification, which achieves high classification accuracy while consuming relatively few resources. Recently, advanced graph convolutional networks (GCNs) [17] and Transformers [18] have gradually gained significant attention. Sun et al. [19] proposed a spectral–spatial fea-

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ture tokenization Transformer (SSFTT) framework to acquire high-level semantic information. However, the high dimensionality of hyperspectral imagery often leads to excessive feature dispersion in ordinary transformers. For this issue, Mei et al. [17] constructed a group-aware hierarchical Transformer (GAHT) that integrates a grouped pixel embedding module to highlight the local connections in the direction of HSI channels. Furthermore, Yang et al. [20] combined the strengths of GCNs and Transformers to create an end-to-end fusion network (GTFN), which demonstrated powerful spatial–spectral feature extraction capability. However, the DL-based techniques mentioned above rely heavily on sufficient labeled samples and expert knowledge, especially in complex forest environments.

Currently, tree species classification under scarce samples mainly involves two strategies. The first strategy involves feature enhancement and data augmentation. Feature enhancement techniques, such as attention mechanisms [21] and feature space reduction (band selection and fusion) [22], can achieve significant improvements in model performance and robustness. The squeeze-and-excitation (SE) module and the convolutional block attention module (CBAM) are highly regarded and widely incorporated into hyperspectral classification models [23], [24]. The normalization-based attention module (NAM) [25] uses a sparse weight penalty to evaluate feature contributions, which endows it with superior simplicity and effectiveness. Wu et al. [1] introduced a CNN-based band selection strategy to reconstruct spectral global relationships by utilizing the most valuable bands of hyperspectral tree species data. However, the feature representations obtained by these methods are learned solely from the original data distribution. We can also enhance sample diversity by employing data augmentation techniques [26]. Basic data augmentation operations such as flipping, cropping, and rotation have become indispensable in addressing the scarce sample issue. Xue et al. [27] designed a spectral–spatial Siamese network (S3Net) to obtain more training samples and features by feeding differentiated patches into the dual branches. Mäyrä et al. [28] used a novel data augmentation technique called mixup to improve tree species classification. Unfortunately, the augmented data may still lack the complexity and variability present in actual forest data, leading to limited accuracy.

The second strategy is to use different learning approaches to deal with limited samples. Common learning paradigms include weakly supervised learning, unsupervised learning, transfer learning, and meta-learning [29]. Unlike weakly supervised and unsupervised paradigms, transfer learning can leverage learned features from label-abundant source domains to data-scarce target domains. Yang et al. [30] first attempted to introduce transfer learning into HSI classification tasks. Subsequently, other models based on transfer learning for plant classification have emerged [31]. However, these models still require sufficient supervision to fine-tune the classification heads. Remarkably, few-shot learning (FSL) provides a viable solution to this issue [32], [33], [34]. FSL serves as a meta-learning approach that can be quickly and efficiently adjusted to novel classes using small labeled instances (usually five samples per class). Compared to one-shot learning [35] and zero-shot learning [36], FSL can better capture the diversity and complexity of ground objects without relying excessively on domain transfer knowledge. Recently, some mainstream FSL-based HSI methods have achieved good performance, such as DCFSL [37], Gia-CFSL [38], CMFSL [39], and ADAFSL [40]. However, these techniques tend to rely on unimodal visual features while neglecting the valuable textual semantics inherent to land cover type labels. To our knowledge, few studies have explored forest tree species classification under few-shot scenarios. Chen et al. [26] described a Proto-MaxUp (PM) framework to mitigate overfitting concerns by improving data augmentation and optimizing the feature extraction backbone for classifying tree species. Furthermore, Chen et al. [32] improved the prototypical network (P-Net) by implementing supervised contrastive learning, which significantly improved the classification accuracy of tree species. However, these methods failed to fully leverage the inference advantages of FSL in complex forest scenes.

With the advancement of pretrained visual language models (VLMs) in visual data exploration and comprehension tasks, transferring pretrained models to downstream remote sensing data analysis tasks has become an essential paradigm [41], [42]. In the field of hyperspectral classification, Zhang et al. [43] proposed a cross-scene HSI network that utilizes the contrastive language–image pretraining (CLIP) model to introduce text features. The substantial prior knowledge of ground objects within these text features significantly enhances visual representation learning. However, this network is limited because the source and target domains must share categories. In short, the remote sensing image interpretation methods based on VLMs are developing vigorously and have become an obvious research trend. However, the exploration of the role of linguistic prior knowledge in fine-grained tree species classification is still in its infancy.

In summary, the current tree species classification utilizing airborne HSIs faces the following pressing issues, which should be immediately addressed.

1) Multispecies mixing and spatial heterogeneity in complex forest scenarios lead to mixed pixel effects and spectral variations, which obscure the key distinguishing features among similar tree species and severely hinder precise tree species identification.

2) Tree species annotation faces severe scarcity of high-quality samples due to expertise barriers and field validation difficulties. Moreover, the existing FSL methods struggle to adapt to domain gaps caused by discrepant taxonomic granularity and forest eco-geographic variations, hindering cross-scene generalization.

3) Unimodal visual representation of various tree species is inadequate and too one-sided, especially for different tree species belonging to the same family and genus, where the spectral characteristics are extremely similar. In fact, the free and rich linguistic semantics contained in category labels are often neglected.

To tackle the aforementioned difficulties, we designed a text-driven few-shot learning (TeDFL) paradigm for forest tree species classification. This framework aims to extract linguistic prior knowledge from category prompts to enrich visual

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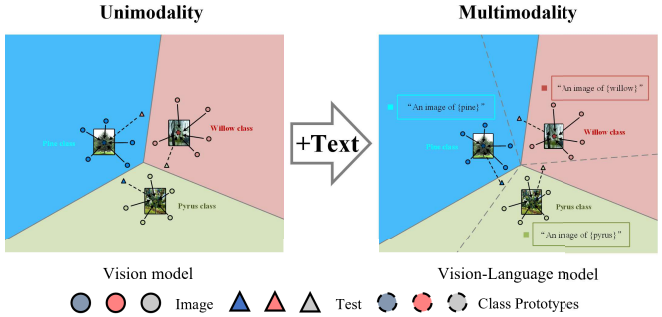


Fig. 1. Illustration of the key concept. Relying solely on single-modal visual features can lead to the misclassification of test samples. Empowered by CLIP, we augment visual features by incorporating text embeddings representing diverse tree species. This approach helps optimize decision boundaries and pull misclassified samples back into the correct class space.

TABLE I
CONFIGURATIONS OF THE VISUAL ENCODER GANET

Layer (type)		Kernel shape	Connected to
Input layer		-	-
CM	Conv3D-BN-ReLU	(3, 3, 3)	Input layer
GAM	Conv3D-BN-ReLU	(1, 1, 1)	CM
	Conv3D-BN-ReLU	(3, 3, 3)	
	Concatenate	-	
	NAM	-	
RM	Conv3D-BN	(1, 1, 1)	CM
	Add	-	
AP	AvgPool3D	(4, 4, 4)	RM
Flatten	-	-	AP

features and improve classification of similar tree species, as illustrated in Fig. 1. Our method primarily comprises two feature extraction branches: visual and textual. Specifically, a ghost attention network (GANet) with ghost attention modules (GAMs) is used as the visual encoder, while the text encoder adopts the CLIP model to introduce rich semantic priors. During training, visual and text features are mapped to a shared-dimensional space. Subsequently, a cross-modal alignment (CMA) strategy is adopted to minimize the differences between visual and text features. Then, we adopt a cross-modal integration (CMI) strategy to fuse multimodal feature embeddings to generate reliable prototypes. The cross-domain FSL mode is used to update all hyperparameters. Experimental results on four realistic forest datasets demonstrate the great potential of VLMs for tree species classification tasks. The main contributions presented in this article are as follows.

1) We develop a TeDFL framework for few-shot forest tree species classification that fully utilizes transferable knowledge from the source domain and enriches visual features with linguistic semantics. This framework achieves precise identification of tree species with similar spectral characteristics using limited labeled target samples.

2) We construct an efficient visual encoder GANet with ghost attention modules to generate highly discriminative visual representations at low computational cost and introduce the CLIP model to acquire semantic priors for enhancing visual feature learning.

3) We design a CMA strategy to align the feature spaces of visual and textual modalities via symmetric contrastive learning. Furthermore, we propose a CMI strategy to obtain multimodal prototype classification constraints, which promotes reliable prototype generation and reduces tree species misclassification.

Following this introduction, the article proceeds as follows. Section II details the study areas and datasets. Section III introduces the TeDFL framework and its components. Section IV provides a comparative evaluation of TeDFL's classification performance. Finally, Section V concludes with future research directions.

II. METHODOLOGY

A. Overall Framework

The overall framework of the proposed TeDFL is illustrated in Fig. 2. Specifically, the TeDFL framework consists of three primary stages: 1) data processing; 2) text-guided meta-learning; and 3) target dataset inference. In the first stage, we first divide the 3-D cube samples from the source and target domains into support and query sets, respectively, and then we map these samples to the same dimensional space to prepare for the subsequent meta-learning stage. In the second stage, we perform text-driven meta-learning to achieve visual feature enhancement through semantic priors. Specifically, this stage includes three key steps: data preparation, feature extraction, and coordination learning. First, we construct text prompts that incorporate category priors for each sample. After that, we utilize the GANet and the pretrained CLIP model, respectively, to extract rich visual features from each category sample and textual features from its corresponding prompt template. Subsequently, the output multimodal features are passed through a projection layer and fed into the CMA module to achieve feature space alignment via symmetric contrastive learning. Meanwhile, we generate and fuse multimodal support prototypes via the CMI strategy to enhance prototype reliability. Furthermore, we retain the raw few-shot classification loss that measures the Euclidean distance (ED) between the support class prototypes $(\mathcal{P}_1, \mathcal{P}_2, \ldots, \mathcal{P}_c)$ and the query features q_i. In the final stage, we use the trained visual encoder GANet and prototype calculation to conduct inference on the target dataset and obtain pixel-level classification results.

B. Visual Encoder Based on the Ghost Attention Module

1) Ghost Attention Module for Discriminative Features at Low Cost: The excessive spectral bands in HSI patches $\mathbf{X} \in \mathbb{R}^{S\times S\times D}$ lead to computational complexity and inefficiency in traditional 3-D convolution operations. To overcome this problem, we introduce the "ghost" concept from Ghostnet to generate multiple similar feature map pairs at low cost [44]. As shown in Fig. 3(a), the first step is to compress the channel dimensionality N of the input feature $\mathbf{X}' \in \mathbb{R}^{S\times S\times D\times N}$ using a 3-D filter $\mathbf{f} \in \mathbb{R}^{k\times k\times D\times N/r}$, where k and r represent the kernel size and scaling factor, respectively. We represent the process as follows:

$$\mathbf{Y} = \mathbf{X}' * \mathbf{f} + \mathbf{b} \tag{1}$$

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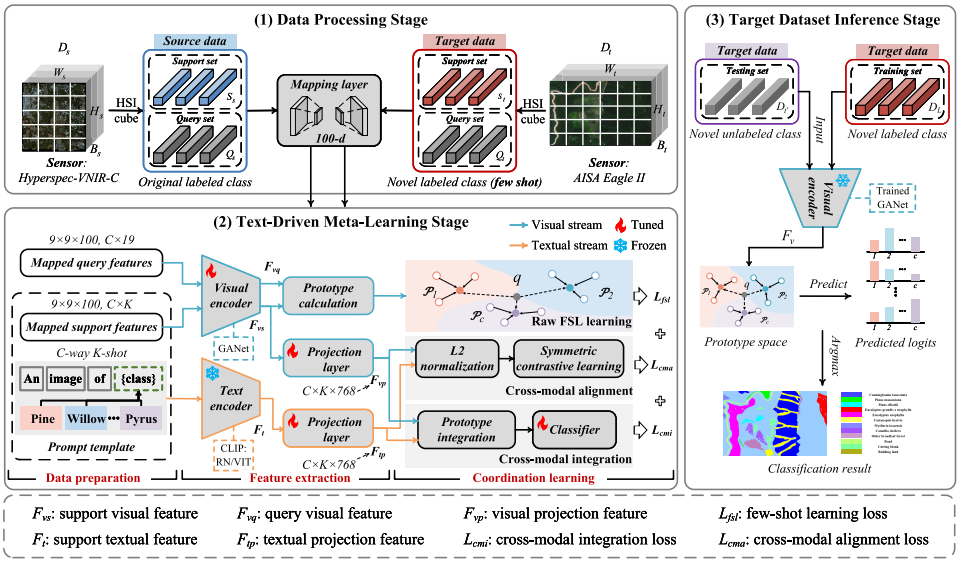


Fig. 2. Overall framework of TeDFL. It consists of three primary stages: data processing, text-guided meta-learning, and target dataset inference. In the text-driven meta-learning stage, the GANet and CLIP extract visual and textual features from samples and their corresponding prompt templates, respectively. These features are projected to the same dimension and then input into the CMA and CMI for multimodal feature alignment and reliable prototype generation, respectively.

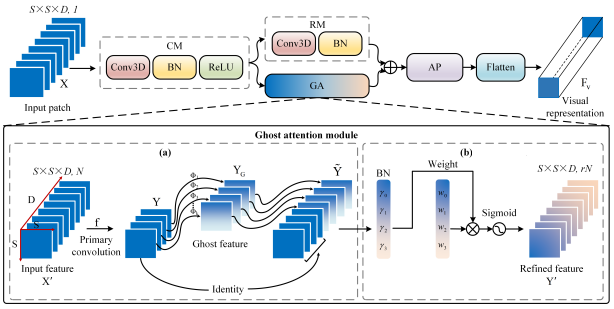


Fig. 3. Network architecture of our visual encoder GANet is mainly composed of the ghost attention module (GAM), which is constructed by connecting the tail of the ghost block to the head of the NAM block.

where $\mathbf{Y} \in \mathbb{R}^{S\times S\times D\times N/r}$ indicates the output intrinsic feature map and the $\mathbf{b}$ is the bias. The second step is to apply inexpensive linear operations to each intrinsic feature in $\mathbf{Y}$ to produce g ghost feature maps. Finally, we obtain the output features $\tilde{\mathbf{Y}}$ by merging the intrinsic feature maps $\mathbf{Y}$ with the ghost feature maps $\mathbf{Y_G}$. The complete procedure can be illustrated as follows:

$$\mathbf{Y_G} = \left[\mathbf{\Phi}_{i,j}\left(\mathbf{y'}_i\right)\right] \quad \forall i = 1,\ldots,N/r \;\; j = 1,\ldots,g \tag{2}$$

$$\tilde{\mathbf{Y}} = \mathrm{Cat}\left[\mathbf{Y},\mathbf{Y_G}\right] \tag{3}$$

where $\mathbf{y'}_i$ denotes the ith original feature map within $\tilde{\mathbf{Y}}$ and $\mathbf{\Phi}_{i,j}$ denotes the jth similar feature map from the ith original feature map. Furthermore, the linear function $\mathbf{\Phi}$ requires less computational effort than the original convolution.

As follows indicates, NAM can directly use batch normalization scale factors to measure channel variances and emphasize importance without additional parameters:

$$B_{\mathrm{out}} = \mathrm{BN}\left(B_{\mathrm{in}}\right) = \gamma\frac{B_{\mathrm{in}} - \mu_\rho}{\sigma_\rho^2} + \beta \tag{4}$$

where μ_ρ and σ_ρ represent the average and standard deviation of a small batch ρ, while γ and β are the trainable parameters used for scaling and shifting. Fig. 3(b) shows the NAM architecture, and the enhanced feature Y' is defined as

$$Y' = \mathrm{Sigmoid}\left(W_\gamma\left(\mathrm{BN}\left(\tilde{\mathbf{Y}}\right)\right)\right) \tag{5}$$

where $W_\gamma = \gamma_i/\sum_{j=0}\gamma_j$ represents the calculated weights. Finally, the final GAM is generated through the collaboration of the ghost block and the NAM block.

2) Proposed GANet Architecture for Exploring Visual Features: The GANet framework, shown in Fig. 3, combines key components like the 3-D convolutional module (CM), the residual module (RM), GAM, an average pooling (AP) layer, and the flatten layer. Table I provides a detailed configuration of the GANet. To facilitate network training, we treat the source dataset $\mathbf{X}_{\mathrm{ori}} \in \mathbb{R}^{H\times W\times B}$ as 3-D cubes with an $S \times S$ window size. Through the GANet, we can effectively extract rich and discriminative embedding features from each cube with low loss. To resolve potential band number differences between data domains, we add a mapping layer before the GANet to consistently project bands into D-dimensions. Importantly, the effectiveness of the GANet is strongly linked to the scaling factor r of GAM, which is extensively examined in Section IV.

C. Text Encoder From the CLIP Model

The CLIP model was trained on 400 million image–text pairs via contrastive learning, which equipped its text encoder with strong and accurate semantic attribute understanding capabilities [45]. This enables the text encoder to adapt flexibly to new downstream tasks, even with limited task-specific data. As described in [45], the text encoder is mainly based on a Transformer structure, which is a 63 M-parameter, 12-layer, 512-width model with eight attention heads. In particular, the

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model width scales with the size of the image encoder. The maximum length of input sentences is 76, given a source dataset involving M categories with natural language names $\{C_1, \ldots, C_m\}$ and a target dataset involving M' categories with natural language names $\{C_1, \ldots, C_{m'}\}$. As depicted in Fig. 2, we integrate each category name C_i^s and C_i^t into a customized prompt template H. The process described above can be expressed as follows:

$$F_t^s = \text{TextEncoder}\left(\text{Tokenize}\left(\left[H; C_i^s\right]\right)\right) \tag{6}$$

$$F_t^t = \text{TextEncoder}\left(\text{Tokenize}\left(\left[H; C_i^t\right]\right)\right) \tag{7}$$

where F_t^s and F_t^t represent the text features of the source domain and the target domain, respectively.

D. Cross-Domain FSL Training Paradigm

Meta-learning is applied to FSL across the source and target domains, extracting transferable knowledge from source classes while learning domain-specific embeddings for target classes [37]. Assume that the source dataset is D_s and the target dataset is D_t, where $D_s \bigcap D_t = \emptyset$, indicating no shared classes between the source and target domains. Taking FSL in the source domain as a case, M_s classes are first randomly chosen from the M source-domain classes. Each of these M_s classes contains K labeled samples, which constitute the support set $S_s = \{(x_{\text{is}}, y_{\text{is}})\}_{\text{is}=1}^{M_s \times K}$. Next, randomly select T unlabeled samples from the remaining M_s classes to construct query set $Q_s = \{(x_{js}, y_{js})\}_{js=1}^{M_s \times T}$. The class distribution P_φ^s, for instance, (x_{js}, y_{js}) within Q_s is calculated through

$$P_\varphi^s\left(y_{js} = k | x_{js} \in Q_s\right) = \frac{\exp\left(-d\left(f_\varphi\left(x_{js}\right), \mathcal{P}_k^s\right)\right)}{\sum_{k=1}^{M_s} \exp\left(-d\left(f_\varphi\left(x_{js}\right), \mathcal{P}_k^s\right)\right)} \tag{8}$$

$$\mathcal{P}_k^s = \frac{1}{K} \sum_{x_{\text{is}} \in S_s} f_\varphi\left(x_{\text{is}}\right) \tag{9}$$

where $\mathcal{P}_k^s$ denotes the prototype of the kth class within S_s and f_φ implements feature extraction using parameters φ. $d(,)$ denotes the similarity between embeddings. Consequently, the source domain FSL loss can be mathematically expressed as

$$L_{\text{fsl}}^s = E_{S_s, Q_s}\left[-\sum_{(x,y) \in Q_s} \log P_\varphi^s\left(y_{js} = k | x_{js}\right)\right]. \tag{10}$$

Consistently, the target-domain FSL inherits the source domain's meta-learning paradigm. Thus, the target-domain FSL loss function is as follows:

$$L_{\text{fsl}}^t = E_{S_t, Q_t}\left[-\sum_{(x,y) \in Q_t} \log P_\varphi^t\left(y_{jt} = k | x_{jt}\right)\right] \tag{11}$$

where $S_t = \{(x_{\text{it}}, y_{\text{it}})\}_{\text{it}=1}^{M_t \times K}$ and $Q_t = \{(x_{jt}, y_{jt})\}_{jt=1}^{M_t \times T}$ denote the support and query sets of the target domain, respectively.

Collectively, the overall loss function for cross-domain FSL can be expressed using the following equation:

$$L_{\text{fsl}} = L_{\text{fsl}}^s + L_{\text{fsl}}^t. \tag{12}$$

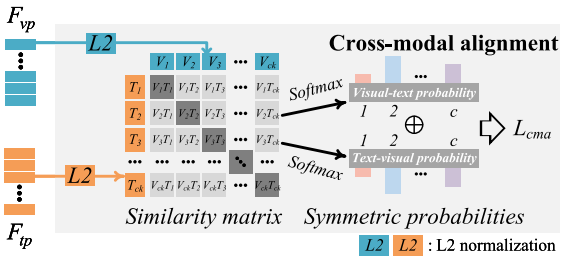


Fig. 4. Structural details of the CMA module. $\{V_1, V_2, V_3, \ldots, V_{ck}\}$ and $\{T_1, T_2, T_3, \ldots, T_{ck}\}$ are the visual embeddings and text embeddings extracted from the L2-normalized visual–text pairs.

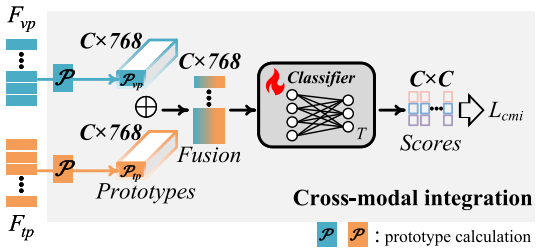


Fig. 5. Structural details of the CMI module. The adaptive classifier with the temperature parameter T enables multimodal prototype classification, enhancing the reliability and robustness of the prototypes.

E. Cross-Modal Adaptation for Enhancing Visual Representations

1) CMA Strategy: Visual features are typically high-dimensional arrays encoding pixel-level information, while text features are usually sparse vectors representing words or tokens [46]. Combining these disparate feature types without alignment can result in degraded model performance. To handle this challenge, we construct a CMA strategy based on contrastive learning, as shown in Fig. 4. Specifically, we first map features from different modalities into a shared embedding space with the same dimension via the projection layer

$$F_{\text{vp}} = \text{VisualProj}\left(F_v\right) \tag{13}$$

$$F_{\text{tp}} = \text{TextProj}\left(F_t\right) \tag{14}$$

where F_v and F_t represent the visual and textual feature representations in the support set, respectively. F_{vp} and F_{tp} denote the output visual and textual results after the projection operation, respectively. Then, we perform L2 normalization on multimodal features for consistent scaling. Finally, we calculate the CMA loss L_{cma} between the visual feature F_{vp} and the text feature F_{tp} using the symmetric contrastive learning, thereby capturing the mutual relationship between the two modalities more comprehensively

$$L_{\text{cma}} = \frac{1}{2}\left(L_{v2t} + L_{t2v}\right) \tag{15}$$

where L_{v2t} (visual-to-text contrastive loss) and L_{t2v} (text-to-visual contrastive loss) are calculated by

$$L_{v2t} = -\frac{1}{N} \sum_{i}^{N} \log \frac{\exp\left(s\left(F_{\text{tp}}^i, F_{\text{vp}}^i\right)\right)}{\sum_{j=1}^{N} \exp\left(s\left(F_{\text{tp}}^i, F_{\text{vp}}^j\right)\right)} \tag{16}$$

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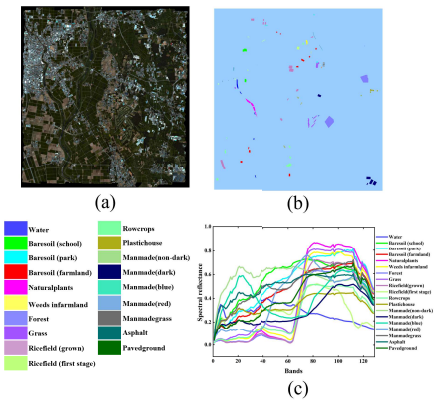


Fig. 6. (a) False-color image, (b) ground-truth map, and (c) mean spectral reflectance for the Chikusei dataset.

TABLE II
COMPREHENSIVE EXPLANATION OF ATTRIBUTES IN THE SOURCE DATASET

Attributes	Chikusei
Spatial size	2517 × 2335
Spectral range (nm)	368-1018
Bands	128
Image resolution (m/pixel)	2.5
Sensor	Hyperspec-VNIR-C
Classes	19

TABLE III
DISTRIBUTION OF SAMPLES PER CLASS IN THE CHIKUSEI DATASET

Label	Color	Class name	Samples
1		Water	2,845
2		Bare soil (school)	2,859
3		Bare soil (park)	286
4		Bare soil (farmland)	4,852
5		Natural plants	4,297
6		Weeds in farmland	1,108
7		Forest	20,516
8		Grass	6,515
9		Rice field (grown)	13,369
10		Rice field (first stage)	1,268
11		Row crops	5,961
12		Plastic house	2,193
13		Manmade (non-dark)	1,220
14		Manmade (dark)	7,664
15		Manmade (blue)	431
16		Manmade (red)	222
17		Manmade grass	1,040
18		Asphalt	801
19		Paved ground	145
		Total	77,592

$$L_{t2v} = -\frac{1}{N}\sum_{i}^{N}\log\frac{\exp\left(s\left(F_{vp}^{i}, F_{tp}^{i}\right)\right)}{\sum_{j=1}^{N}\exp\left(s\left(F_{vp}^{i}, F_{tp}^{j}\right)\right)} \tag{17}$$

where N indicates the number of instances within the support set and $s(F_i, F_j) = (F_i F_j^T)/(\|F_i\|\|F_j\|)$ indicates the cosine similarity (CS).

2) CMI Strategy: Relying solely on a few-shot samples from a single modality can lead to biased or incomplete prototypes. Thus, we propose a CMI strategy that fully leverages the visual and textual prototype representations of support samples to generate robust and reliable class prototypes. As shown in Fig. 5, we first perform prototype calculation on the multimodal projection features. Then, we use an element-wise addition operation to obtain the multimodal prototype. Finally, a simple yet effective adaptive classifier normalizes input features and scales logits via a learnable logit scale to adjust decision boundaries adaptively. Therefore, for the visual prototype $\mathcal{P}_{vp}^{k}$ and textual prototype $\mathcal{P}_{tp}^{k}$ of the kth class sample of S_s, we can obtain the class distribution P_φ formula as follows:

$$P_\varphi(y_i = k) = \frac{\exp\left(c\left(f\left(\mathcal{P}_{vp}^{k}, \mathcal{P}_{tp}^{k}\right)\right)\right)}{\sum_{k=1}^{N}\exp\left(c\left(f\left(\mathcal{P}_{vp}^{k}, \mathcal{P}_{tp}^{k}\right)\right)\right)} \tag{18}$$

where c denotes a classifier with learnable temperature parameter T, consisting of a single linear layer, $f(,)$ represents the add operation, and y signifies the label of the ith sample. Consequently, the CMI source-domain loss L_{cmi}^{s} and CMI target-domain loss L_{cmi}^{t} can be, respectively, formulated as

$$L_{cmi}^{s} = E_s\left[-\sum_{(x,y)\in S_s}\log P_\varphi\left(y_{is} = k|x_{is}\right)\right] \tag{19}$$

$$L_{cmi}^{t} = E_t\left[-\sum_{(x,y)\in S_t}\log P_\varphi\left(y_{it} = k|x_{it}\right)\right]. \tag{20}$$

Note that FSL is conducted alternately on both the source and target domains. Hence, the overall loss function L^s for the source domain can be denoted as follows:

$$L^s = L_{fsl}^{s} + L_{cma}^{s} + L_{cmi}^{s}. \tag{21}$$

Also, the overall loss function L^t for the target domain can be given by the following expression:

$$L^t = L_{fsl}^{t} + L_{cma}^{t} + L_{cmi}^{t}. \tag{22}$$

III. EXPERIMENTS

A. Data Description

In this study, the experimental datasets consist mainly of source and target datasets. Chikusei [47] (base classes) serves as a source dataset with rich label information but coarse categories, while the target dataset uses four fine-grained forest tree species scenes (novel classes) with limited labeled samples, including Gaofeng Forest Farm A and B (GFF-A [48] and GFF-B [16]), Tea Farm [49], and Xiong An [50] scenes. This section will thoroughly introduce the details of each dataset.

TABLE IV
COMPREHENSIVE EXPLANATION OF ATTRIBUTES IN THE TARGET DATASETS

Datasets	GFF-A	GFF-B	Tea Farm	Xiong An
Size (pixels)	1056×914	572×906	348×512	700×800
Range (nm)	400–987	400–987	417–855	400–1000
Bands	125	125	80	256
Resolution (m)	1	1	2.25	0.5
Sensor	AISA Eagle II	AISA Eagle II	APHI	FSMHI
Classes	11	12	10	13

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TABLE V
DISTRIBUTION OF SAMPLES PER-CLASS ACROSS VARIOUS TARGET DATASETS

Label	GFF-A		GFF-B	
	Class name	Samples	Class name	Samples
1	Acacia melanoxylon	119	Cunninghamia lanceolata	80,743
2	Castanopsis hystrix	117	Pinus massoniana	5,340
3	Cunninghamia lanceolata	120	Pinus elliottii	4,175
4	Eucalyptus grandis x urophylla	119	Eucalyptus grandis x urophylla	14,931
5	Eucalyptus urophylla	119	Eucalyptus urophylla	40,275
6	Mytilaria laosensis	115	Castanopsis hystrix	25,461
7	Pinus elliottii	119	Mytilaria laosensis	3,000
8	Pinus massoniana	118	Camellia oleifera	8,324
9	Road	120	Other broadleaf forest	10,741
10	Soft broadleaved	119	Road	10,472
11	Cutting-site	112	Cutting blank	10,170
12			Building land	68
	Total	1,297		213,700
Label	**Tea Farm**		**Xiong An**	
	Class name	**Samples**	**Class name**	**Samples**
1	Masson pine	5,806	Acer negundo	57,096
2	Bamboo forest	2,318	Willow	9,173
3	Tea plant	28,428	Sophora japonica	39,358
4	Reed	214	Fraxinus chinensis	41,343
5	Rice paddy	6,809	Waters	21,007
6	Sweet potato	817	Corn	28,761
7	Carawa	429	Pyrus	86,138
8	Weed	1,861	Soybean	1,303
9	Water	6,141	Poplar	11,064
10	Building and road	911	Vegetable field	6,043
11			Grass	9,292
12			Peach	29,874
13			Building	7,476
	Total	53,734		347,928

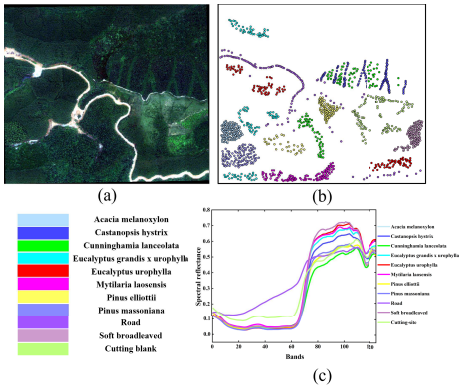


Fig. 7. (a) False-color image, (b) ground-truth map, and (c) mean spectral reflectance for the GFF-A dataset.

1) Source-Domain Dataset: The Chikusei[1] dataset was gathered in 2014 using the hyperspectral visible/near-infrared C series sensor (Hyperspec-VNIR-C) in Chikusei, Japan (140°0′30″E, 36°17′41″N). This dataset comprises a spatial coverage of 2517 × 2335 pixels with 19 categories, offering rich data for training models across different domains. Table II provides detailed attribute information, while Table III displays the sample distribution per class in the Chikusei dataset. Fig. 6 illustrates the false-color image, ground-truth map, and mean spectral reflectance associated with the Chikusei dataset.

2) Target-Domain Dataset: The GFF-A,[2] GFF-B,[3] Tea Farm,[4] and Xiong An[5] datasets primarily focus on forest

[1]Chikusei: https://naotoyokoya.com/Download.html
[2]GFF-A: data will be made available on request.
[3]GFF-B: https://github.com/longbaochenlong/SCL-P-Net
[4]Tea Farm: https://www.geodoi.ac.cn/WebCn/doi.aspx?Id=720
[5]Xiong An: https://www.geodoi.ac.cn/doi.aspx?Id=1687

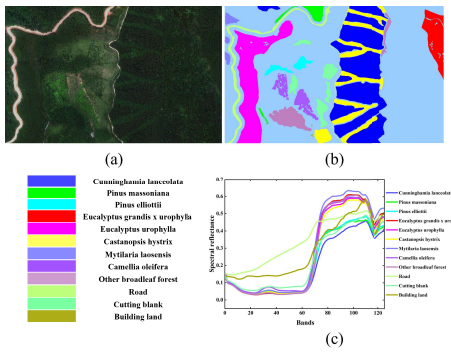


Fig. 8. (a) False-color image, (b) ground-truth map, and (c) mean spectral reflectance for the GFF-B dataset.

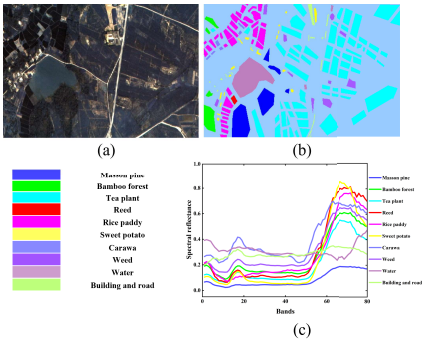


Fig. 9. (a) False-color image, (b) ground-truth map, and (c) mean spectral reflectance for the Tea Farm dataset.

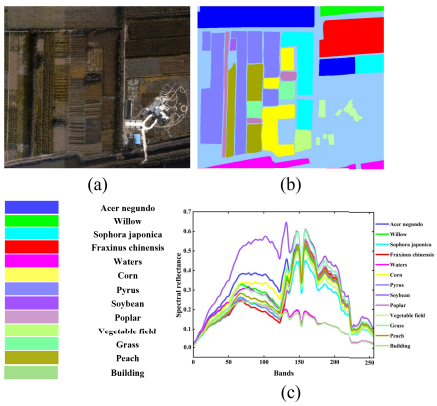


Fig. 10. (a) False-color image, (b) ground-truth map, and (c) mean spectral reflectance for the Xiong An dataset.

scenes and include diverse tree species. Table IV provides comprehensive descriptions of the four forest datasets, while Table V presents the per-class sample distribution across these datasets.

The GFF-A dataset was collected by the AISA Eagle II in 2018 over Nanning, Guangxi, China (108°22′15″E, 22°57′45″N). It has 1,297 samples across 11 classes, including 9 tree species categories. Fig. 7 shows the visualization maps of the GFF-A dataset.

The GFF-B dataset, as a derivative dataset of GFF-A, has more comprehensive annotations. The dataset comprises 2 13 700 samples distributed across 12 categories, including 9

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TABLE VI

CLASSIFICATION ACCURACY OF VARIOUS ALGORITHMS ON THE GFF-A DATASET

Label	Non-FSL-based methods					FSL-based methods							
	3DDL	SSFTT	S3Net	GAHT	GTFN	DCFSL	HFSL	Gia-CFSL	CMFSL	RCA-PDA	STBDIP	ADAFSL	TeDFL
1	37.54±9.10	42.45±10.20	71.28±14.57	**81.92±9.27**	80.52±8.61	75.43±10.54	68.96±16.31	74.56±12.46	72.80±5.74	66.67±19.09	70.27±23.69	81.57±6.68	71.84±8.66
2	21.88±6.60	25.98±6.28	75.98±13.91	44.55±11.60	68.21±5.91	64.28±13.53	71.05±19.61	68.75±10.81	77.67±10.00	62.50±10.67	64.22±16.44	59.82±11.26	**80.45±7.32**
3	59.39±10.13	52.69±13.14	66.36±10.03	61.13±8.97	75.82±2.82	58.26±10.32	46.15±18.50	73.91±11.81	61.73±11.23	79.13±12.55	77.67±11.10	72.17±8.84	**80.00±8.35**
4	28.60±10.90	55.17±9.70	56.23±21.07	53.50±15.40	82.80±5.28	63.15±13.78	25.86±14.37	37.71±14.15	**89.47±16.85**	32.45±15.35	64.86±19.77	53.50±13.50	78.51±13.99
5	25.61±5.46	28.24±8.07	45.59±14.05	37.45±7.09	52.10±6.53	27.19±11.44	56.03±18.04	67.54±11.20	28.94±11.63	71.92±16.73	36.03±15.52	50.87±13.92	**72.46±12.64**
6	33.36±8.40	37.18±13.72	63.33±12.26	49.90±13.90	69.63±8.32	65.45±12.86	62.50±12.50	63.63±9.58	**82.72±9.81**	67.27±12.74	71.96±9.68	63.63±12.05	75.82±3.33
7	28.60±8.92	42.10±8.83	68.25±8.22	50.26±16.58	62.45±2.31	49.12±12.28	**89.65±21.63**	69.29±9.51	65.78±8.61	70.17±10.70	44.14±19.62	70.17±8.30	68.33±7.54
8	36.73±6.86	54.42±13.53	76.94±10.97	81.15±11.31	87.25±3.77	75.22±11.69	75.65±21.76	67.25±9.30	73.45±10.25	78.76±7.56	75.45±15.50	83.18±9.45	**87.61±4.89**
9	77.65±16.34	93.21±5.28	96.27±3.90	92.86±6.00	92.00±1.68	**97.39±12.34**	36.75±18.57	93.91±5.70	94.78±5.28	92.17±5.34	97.32±1.70	87.82±3.01	89.13±4.58
10	65.35±12.08	83.94±8.58	**88.71±7.25**	86.66±8.52	78.07±1.84	78.94±6.54	81.03±24.30	73.68±7.05	88.59±6.76	78.94±2.86	82.88±12.25	70.17±6.42	87.37±3.49
11	92.43±6.64	98.97±1.35	97.54±3.85	94.39±4.15	97.75±1.26	92.52±2.08	99.08±3.23	97.19±2.98	93.45±2.08	96.26±2.38	**100.00±6.54**	97.19±1.43	96.64±1.68
OA(%)	45.97±3.40	55.75±2.04	73.21±2.15	66.65±2.32	76.89±2.31	67.79±4.73	64.55±7.79	71.49±3.35	75.28±3.01	72.30±2.72	71.21±2.23	71.73±2.44	**80.67±1.86**
AA(%)	46.10±3.36	55.85±2.03	73.32±2.11	66.71±2.32	76.96±2.32	67.90±4.71	64.79±7.75	71.58±3.34	75.40±2.99	72.39±2.71	71.35±2.21	71.83±2.45	**80.74±1.85**
Kappa	40.57±3.73	51.32±2.24	70.53±2.36	63.31±2.55	74.58±2.54	64.57±5.21	61.02±8.57	68.64±3.68	72.81±3.30	69.53±2.99	68.34±2.45	68.91±2.68	**78.73±2.04**

TABLE VII

CLASSIFICATION ACCURACY OF VARIOUS ALGORITHMS ON THE GFF-B DATASET

Label	Non-FSL-based methods					FSL-based methods							
	3DDL	SSFTT	S3Net	GAHT	GTFN	DCFSL	HFSL	Gia-CFSL	CMFSL	RCA-PDA	STBDIP	ADAFSL	TeDFL
1	64.72±10.88	51.97±12.16	66.08±20.45	65.76±16.87	76.57±5.33	75.40±9.85	68.03±17.37	82.81±5.14	81.59±5.60	75.84±6.19	**85.88±5.75**	78.86±3.07	85.01±4.41
2	43.15±13.90	**85.81±2.60**	75.33±6.98	65.76±7.57	82.84±1.15	58.74±8.23	62.69±14.16	60.49±11.11	70.38±10.89	67.84±14.20	52.33±16.58	75.41±4.49	74.82±7.89
3	52.62±9.84	89.11±6.00	88.94±5.07	71.58±14.04	**89.21±3.31**	67.58±14.39	80.49±9.22	73.39±7.22	77.75±6.24	81.17±12.65	65.78±19.95	80.06±3.17	88.35±4.80
4	56.52±14.29	53.76±6.34	83.32±9.63	77.96±10.76	62.00±6.02	61.38±17.30	64.57±22.07	77.74±7.39	**84.62±4.79**	76.14±10.03	52.51±19.36	71.02±12.89	83.38±6.91
5	43.71±12.96	47.82±3.83	62.81±17.05	29.07±8.75	51.06±7.92	55.72±6.66	58.30±24.82	59.38±6.41	61.68±10.04	69.99±10.30	65.17±12.53	61.31±13.02	**82.37±6.43**
6	35.17±10.21	58.64±6.51	54.77±18.15	28.75±6.90	50.08±6.55	44.07±15.87	42.50±18.74	46.19±12.12	57.30±8.13	57.46±12.05	44.03±24.42	64.33±4.94	**73.34±3.86**
7	69.52±16.45	**96.63±2.03**	71.24±18.63	75.92±11.57	79.00±6.44	84.76±6.34	78.57±15.98	81.28±9.50	90.38±4.61	88.68±7.96	90.07±4.75	91.81±2.05	94.86±3.16
8	68.98±13.72	**95.46±10.78**	82.19±9.54	89.29±6.22	90.40±3.88	86.55±6.49	85.65±7.80	82.25±9.05	89.50±3.74	79.14±10.34	81.07±7.60	81.36±1.88	89.70±5.54
9	57.70±13.68	**88.25±2.23**	86.19±4.42	76.81±14.88	83.16±3.64	74.69±4.22	75.43±16.75	73.30±7.48	74.56±6.22	69.77±7.33	74.45±10.06	71.49±14.62	77.82±7.31
10	86.72±6.05	95.61±0.77	95.01±2.38	94.20±1.76	**96.77±0.71**	92.96±3.41	93.57±4.78	91.04±6.24	94.18±2.81	92.95±2.56	93.75±3.14	92.47±2.68	92.53±4.25
11	87.57±8.00	87.58±3.49	86.56±9.57	91.52±5.85	95.22±2.65	96.32±3.16	**99.66±0.55**	96.12±3.85	96.19±4.16	92.14±5.84	94.20±5.39	98.38±0.71	97.85±2.47
12	96.98±4.04	98.79±13.27	**100.00±0.00**	98.10±5.71	**100.00±0.00**	96.03±10.86	96.03±9.97	95.56±7.16	97.94±5.68	95.24±10.86	94.13±11.64	87.83±15.08	94.76±11.45
OA(%)	57.94±6.58	61.67±5.18	70.11±7.71	59.65±6.77	70.78±2.82	68.80±4.94	66.99±9.90	73.45±2.28	76.19±2.45	74.09±2.24	73.51±3.08	74.72±2.27	**83.76±1.86**
AA(%)	63.61±5.37	79.23±2.93	79.37±2.64	72.08±2.58	79.69±0.77	74.52±4.20	75.46±5.73	76.63±1.63	81.34±1.18	78.86±2.94	74.45±4.90	79.53±4.24	**86.23±1.80**
Kappa	50.22±7.16	55.81±5.14	64.48±7.95	52.75±6.61	64.75±2.99	62.20±5.43	60.64±10.88	67.47±2.44	70.84±2.79	68.45±2.53	67.14±3.72	69.05±2.56	**79.93±2.20**

tree species categories. Fig. 8 displays the visualization maps of the GFF-B dataset.

The Tea Farm dataset was collected using an Aerosol Pushbroom Hyperspectral Imager (APHI) in 2017 over Changzhou, Jiangsu, China (119°22′50″E, 31°40′40″N). It encompasses 53 734 samples across 10 classes, including 7 tree species categories. Fig. 9 displays the visualization maps for the Tea Farm dataset.

The Xiong An dataset was collected utilizing the Full Spectrum Multimodal Hyperspectral Imager (FSMHI) in 2019 over Xiongan, Hebei, China (111°33′15″E, 38°57′0″N). This dataset encompasses 3 47 928 samples across 13 classes, including 7 tree species categories. Further details on visual representations are provided in Fig. 10.

B. Experimental Setting

1) Implementation Details: To gain transferable knowledge, 200 annotated samples per class were chosen from the base Chikusei dataset. During the training process, $L = 5$ annotated samples per class were chosen at random from the target datasets. The input hyperspectral patches were set with a uniform window size S and number of bands D at 9 and 100, respectively. We set the 3-D convolution filters N to 8 and the scaling coefficient r to 2. The HSI classification was formulated as C-way, K-shot episodes in our FSL framework, where C denotes class count and K indicates support samples per category. C corresponded to target dataset classes (11 for GFF-A, 12 for GFF-B, 10 for Tea Farm, and 13 for Xiong An), while K was set to 1 to emulate the FSL environment. In every episode, 19 random samples were chosen from the remaining C class instances to establish unlabeled query samples. We employed the Adam optimizer with a learning rate (lr) of 0.01, cross-entropy loss, and 10 000 training iterations to achieve efficient parameter optimization. For quantitative evaluation of the performance of visual features enhanced by text priors, we used a fixed text template: "An image of {class name}," consistent with the description style in CLIP pretraining.

All training processes were executed on an NVIDIA Titan RTX 3090 GPU server, and the major software setup included Python 3.9.7 and PyTorch 1.12.0.

2) Comparison Algorithms and Evaluation Metrics: In this study, 12 mainstream methods were employed for comparison with the proposed method, TeDFL. Among these, five non-FSL-based methods were involved, such as 3DDL [15], S3Net [27], GTFN [20], SSFTT [19], and GAHT [17], which are based on CNN, GCN, and Transformer architectures. Additionally, seven advanced FSL-based methods were selected for comparison, including DCFSL [37], HFSL [51], Gia-CFSL [38], CMFSL [39], RCA-PDA [52], STBDIP [53], and ADAFSL [40]. All comparison methods followed the default values from the open-source codes and carefully further tuned the hyperparameters to achieve optimal performance. To ensure fair evaluation, we unified the input patch size and random sampling factors of all methods, and retained the mean and standard deviation from 10 trials. Three essential metrics are selected to evaluate models: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (100% × Kappa).

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TABLE VIII
CLASSIFICATION ACCURACY OF VARIOUS ALGORITHMS ON THE TEA FARM DATASET

Label	Non-FSL-based methods					FSL-based methods							
	3DDL	SSFTT	S3Net	GAHT	GTFN	DCFSL	HFSL	Gia-CFSL	CMFSL	RCA-PDA	STBDIP	ADAFSL	TeDFL
1	89.76±2.41	85.47±6.66	91.76±6.83	93.46±4.76	92.72±4.23	93.48±2.93	92.29±3.74	92.38±2.98	93.76±3.83	92.55±3.59	**94.47±1.80**	93.25±3.86	93.69±3.04
2	46.79±5.78	73.93±5.98	62.20±10.70	79.66±4.48	60.55±2.73	83.36±4.91	88.02±7.52	81.63±7.43	85.23±5.74	80.56±10.45	80.61±7.45	75.40±7.81	**88.78±6.81**
3	81.38±9.34	87.93±5.88	86.37±8.72	88.75±5.35	85.15±4.93	90.69±4.43	89.99±6.34	**91.60±3.00**	90.52±4.68	90.21±4.58	90.85±3.47	91.42±4.45	91.50±2.85
4	99.57±0.87	99.81±0.44	97.32±4.18	**100.00±0.00**	99.81±0.23	**100.00±0.00**	**100.00±0.00**	99.38±1.03	99.86±0.43	99.86±0.43	99.62±0.77	**100.00±0.00**	99.90±0.29
5	98.26±0.79	ul 9.22±0.43	97.87±1.78	98.71±0.71	98.91±0.39	98.81±1.19	96.52±3.03	97.92±2.59	98.95±0.79	97.23±1.42	**99.38±0.58**	96.67±3.12	99.08±0.66
6	91.48±7.03	96.08±4.89	94.83±3.00	93.21±6.05	81.65±5.72	92.89±6.24	**97.07±2.98**	92.08±7.40	96.28±3.36	85.90±6.70	90.91±11.18	89.37±4.30	93.72±6.56
7	90.28±4.45	86.65±7.55	97.55±6.29	96.27±4.64	98.82±1.31	97.17±4.07	96.49±7.57	97.26±3.26	96.86±4.17	92.62±8.91	**99.79±0.25**	98.90±0.95	99.27±1.07
8	39.06±10.11	56.23±11.99	68.01±13.87	62.87±4.47	**81.06±8.38**	65.40±11.93	69.85±14.29	68.43±8.52	70.70±6.48	71.10±10.06	67.24±11.14	74.53±8.14	80.24±8.38
9	98.97±0.34	**99.71±0.23**	98.47±0.88	99.02±0.41	97.44±1.42	98.98±0.36	98.11±0.80	98.89±0.36	98.81±0.97	99.08±0.46	99.00±0.40	99.21±0.68	98.96±0.36
10	94.07±4.83	96.62±4.75	97.92±2.10	97.83±2.82	93.02±3.67	98.32±2.99	98.91±1.47	99.30±0.74	99.16±1.08	96.25±5.05	**99.93±0.13**	98.71±1.39	98.38±3.95
OA(%)	83.99±5.11	89.05±3.38	88.57±4.62	90.73±3.13	88.16±2.60	92.13±2.59	91.56±2.95	92.20±1.94	92.29±2.56	91.39±2.74	92.25±1.92	92.09±2.15	**93.29±1.67**
AA(%)	82.96±1.36	88.17±2.61	89.23±2.40	90.98±1.18	88.91±1.28	92.01±1.64	92.72±1.25	91.89±1.63	92.01±1.29	90.54±2.95	92.18±1.80	91.75±1.78	**94.35±1.59**
Kappa	78.02±5.97	84.58±4.42	84.06±5.81	86.91±4.07	83.37±3.38	88.79±3.46	88.02±3.89	88.96±2.65	89.03±3.44	87.74±3.67	88.92±2.59	88.68±2.90	**90.39±2.29**

TABLE IX
CLASSIFICATION ACCURACY OF VARIOUS ALGORITHMS ON THE XIONG AN DATASET

Label	Non-FSL-based methods					FSL-based methods							
	3DDL	SSFTT	S3Net	GAHT	GTFN	DCFSL	HFSL	Gia-CFSL	CMFSL	RCA-PDA	STBDIP	ADAFSL	TeDFL
1	72.31±9.75	80.00±6.54	64.21±14.80	67.15±7.45	54.15±12.08	68.92±11.13	67.01±15.55	69.57±8.64	76.38±7.32	72.07±14.69	66.81±10.74	75.40±1.47	**81.69±6.56**
2	66.23±10.43	94.47±3.41	97.60±0.23	74.64±15.59	97.64±0.94	95.45±2.83	66.52±13.62	94.16±3.25	93.73±5.44	89.88±8.90	93.58±9.40	85.63±11.36	**98.57±1.42**
3	48.85±17.23	69.05±14.39	73.96±6.75	51.64±10.62	75.36±3.60	75.96±9.79	50.02±14.98	72.61±11.32	74.70±12.51	75.21±9.27	75.89±9.11	**83.41±3.39**	79.79±7.68
4	86.31±5.18	93.21±4.08	**98.44±0.56**	84.65±7.58	94.55±3.64	95.36±2.83	89.55±9.89	92.75±4.42	95.74±2.73	93.88±2.81	94.78±2.97	98.22±0.15	97.65±0.97
5	87.80±7.37	90.74±5.68	90.36±3.81	87.77±6.97	**94.49±1.97**	81.78±13.28	81.67±9.10	80.01±10.35	82.51±10.31	82.80±9.51	87.89±5.07	87.59±4.83	87.12±7.02
6	43.08±12.55	63.46±13.48	**82.24±6.31**	63.06±10.19	69.24±8.81	75.96±11.96	69.15±6.36	71.81±15.31	78.96±8.79	66.36±11.36	70.88±11.09	75.36±5.58	77.43±11.13
7	25.95±9.85	39.32±14.00	38.76±8.63	32.69±13.91	44.72±12.07	52.67±8.95	46.55±15.39	47.51±12.11	58.50±8.11	53.01±9.08	51.24±10.71	41.06±2.28	**64.04±6.79**
8	86.12±2.62	90.74±4.46	**96.96±2.12**	89.75±5.55	96.46±1.87	88.24±7.88	89.33±4.68	85.17±6.75	88.60±5.23	86.23±6.42	93.19±3.43	91.29±1.34	88.72±9.25
9	58.04±11.29	77.80±11.41	79.91±10.16	67.11±12.89	86.79±6.70	77.77±4.19	71.73±7.63	70.38±12.83	80.23±6.70	70.68±8.54	80.15±7.26	**88.92±1.52**	82.77±7.25
10	36.90±7.30	59.48±10.77	67.12±9.67	28.25±7.35	79.75±7.90	71.71±9.44	69.00±18.67	68.56±10.84	71.64±12.95	77.91±8.37	68.64±9.58	62.08±6.58	**80.26±10.55**
11	53.43±15.09	63.14±14.13	60.68±22.81	46.64±14.49	**76.00±7.45**	64.93±8.65	44.93±7.45	62.40±7.84	63.98±6.35	59.74±13.00	69.68±8.43	34.85±2.89	68.99±9.35
12	30.66±7.23	50.21±10.88	44.63±0.91	36.30±12.31	**65.44±9.82**	41.88±9.20	48.05±3.17	48.45±6.61	46.62±8.30	36.82±11.27	45.78±7.42	34.42±8.25	55.44±7.21
13	82.70±9.20	91.93±7.23	90.34±2.68	88.04±10.28	95.16±2.06	79.78±12.75	88.16±3.83	82.85±13.66	84.20±8.37	81.85±14.33	88.23±7.81	**97.42±0.26**	86.27±11.02
OA(%)	53.33±5.04	66.78±5.41	66.49±5.24	56.71±3.89	67.92±2.61	69.10±3.43	62.26±0.47	67.02±5.53	72.47±3.56	67.91±3.65	68.91±3.18	67.65±1.50	**76.96±2.75**
AA(%)	59.88±3.33	74.12±3.44	75.79±4.79	62.90±2.38	79.21±1.58	74.s65±3.03	67.82±1.58	72.79±3.39	76.60±2.53	72.80±3.42	75.90±2.64	73.51±1.48	**80.67±1.49**
Kappa	47.95±5.23	62.73±5.84	62.36±5.76	51.68±3.98	64.25±2.68	65.08±3.82	57.29±0.55	62.79±5.98	68.70±3.91	63.55±4.28	64.92±3.45	63.79±1.70	**73.76±3.03**

C. Classification Performance Comparison

In this section, we leverage four realistic forest scenarios to conduct classification experiments: GFF-A, GFF-B, Tea Farm, and Xiong An. Classification accuracies across four mainstream datasets (GFF-A, GFF-B, Tea Farm, and Xiong An) are reported with distinct formatting: **bold** for top performance, underlined for runner-up.

1) Quantitative Comparisons: Tables VI–IX present the quantitative results of diverse classification algorithms on the GFF-A, GFF-B, Tea Farm, and Xiong An datasets, respectively. Notably, the proposed TeDFL method exhibits consistent superiority across four forest scene datasets. S3Net explored rich spectral features via multiscale patches, surpassing the 3DDL baseline in accuracy. With advanced backbone networks, SSFTT, GAHT, and GTFN enhance tree species classification, with the GTFN achieving 76.89% OA on the GFF-A dataset. The FSL paradigm can obtain robust representations and transferable meta-information, which significantly enhances precision in tree species identification. HFSL performs FSL to acquire transferable knowledge from natural images, but it fails to consider the differences between domains. DCFSL first considers the cross-domain issue, which leads to higher classification accuracy. Following DCFSL, Gia-CFSL, RCA-PDA, STBDIP, and ADAFSL were subsequently proposed. However, their performance exhibits instability across diverse forest scenarios. CMFSL integrated the Mahalanobis distance and achieved consistently high classification accuracy. TeDFL introduces text semantic priors to optimize the FSL process. Compared to the advanced CMFSL, TeDFL improves the OA by 5.39%, 7.57%, 1.00%, and 4.49% on the GFF-A, GFF-B, Tea Farm, and Xiong An datasets, respectively. For classes that are prone to misclassification by comparative methods, such as *Eucalyptus urophylla* and *Castanopsis hystrix* from the GFF-A and GFF-B datasets, *Weeds* from the Tea Farm dataset, *Pyrus*, *Vegetable field*, and *Peach* from the Xiong An dataset, the TeDFL method achieves robust recognition performance with consistently high accuracy.

2) Qualitative Comparisons: Figs. 11–14 present the visualization results of various models across the GFF-A, GFF-B, Tea Farm, and Xiong An datasets, respectively. Due to the complexity of actual forest scenes in GFF-A and GFF-B datasets, some methods have serious misclassification problems (such as 3DDL and DCFSL). Owing to the relatively concentrated category distribution and simpler classification system in the Tea Farm dataset, most of the models can achieve relatively high accuracy. For the Xiong An dataset with a complex classification system, all models except CMFSL have more misclassified points. As illustrated in Figs. 11–14, TeDFL demonstrates substantial performance advantages with fewer misclassified pixels, even in challenging categories.

IV. DISCUSSIONS

A. Investigation of the Parameter Sensitivity

In this section, we primarily discuss and analyze the sensitivity of two parameters: the scaling factor r of the GAM and the learning rate lr during training.

1) Scaling Factor r: The scaling factor r in GAM represents the degree of data dimension compression; in particular,

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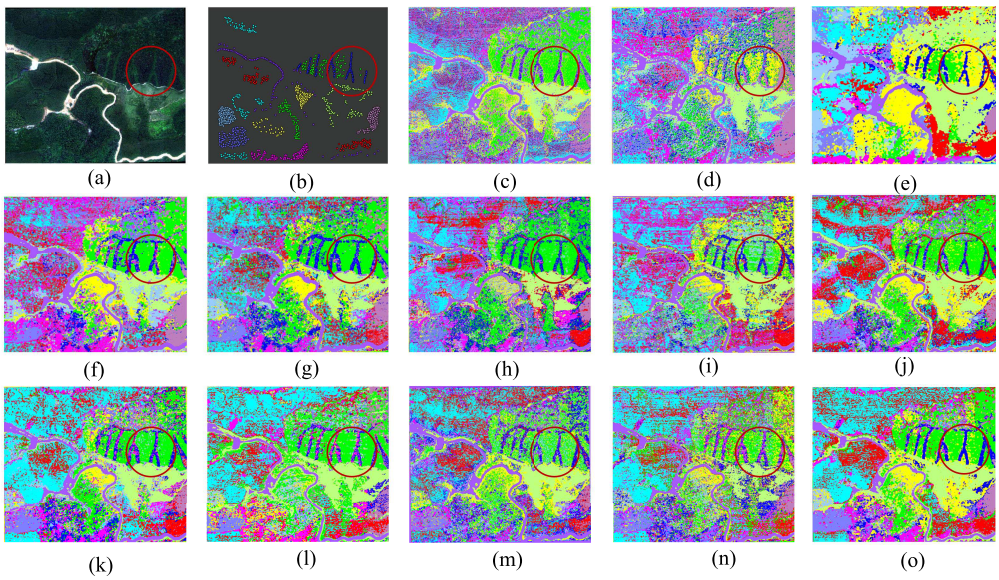


Fig. 11. Visual representations of various classification algorithms on the GFF-A dataset. (a) False-color image. (b) Ground truth. (c) 3DDL. (d) SSFTT. (e) S3Net. (f) GAHT. (g) GTFN. (h) DCFSL. (i) HFSL. (j) Gia-CFSL. (k) CMFSL. (l) RCA-PDA. (m) STBDIP. (n) ADAFSL. (o) TeFL.

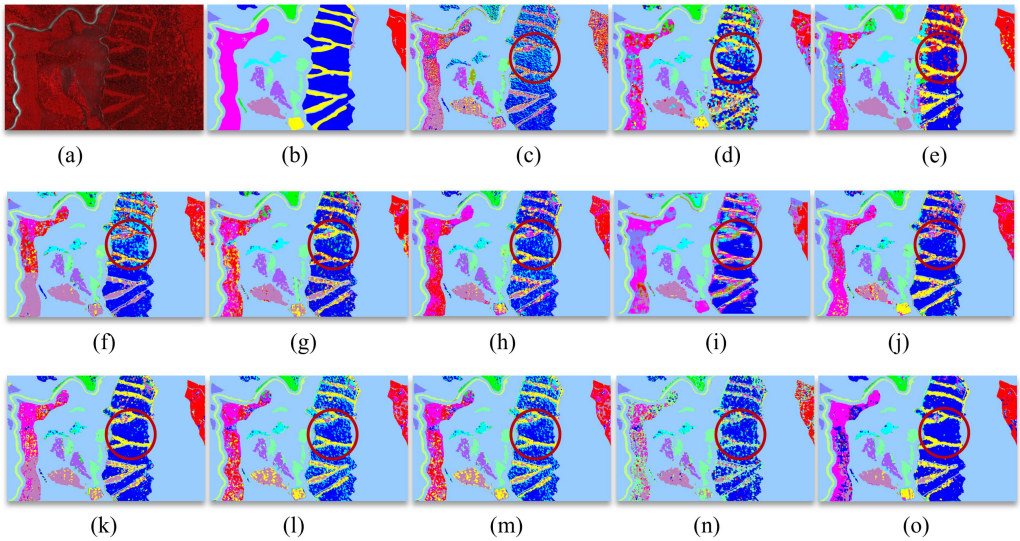


Fig. 12. Visual representations of various classification algorithms on the GFF-B dataset. (a) False-color image. (b) Ground truth. (c) 3DDL. (d) SSFTT. (e) S3Net. (f) GAHT. (g) GTFN. (h) DCFSL. (i) HFSL. (j) Gia-CFSL. (k) CMFSL. (l) RCA-PDA. (m) STBDIP. (n) ADAFSL. (o) TeFL.

a larger r leads to greater model compression and fewer parameters. We set r within the range of $\{2, 4, 6, 8\}$, and the box plot in Fig. 15 illustrates the trends of changes in r across different datasets. As r increases, the OA on the GFF-B dataset exhibits a tendency of first decreasing and then increasing, achieving higher classification performance when $r = 2$. On the GFF-A and Tea Farm datasets, classification accuracy first rises then falls as r increases, with OA peaking at $r = 4$. Additionally, on the Xiong An dataset, the increase in r is accompanied by a slow decrease in OA. Overall, r is set to 2 for both the GFF-B and Xiong An datasets, and r is set to 4 for the GFF-A and Tea Farm datasets.

2) Learning Rate Lr: As a critical hyperparameter in DL, the learning rate lr determines the degree of weight and bias adjustments in each iteration of the gradient descent algorithm. Learning rates that are too large or too small can lead to unstable model training, slow convergence, or poor generalization capabilities. Therefore, we conducted five sets of experiments on each dataset, with the learning rate lr ranging from $\{1e-1, 1e-2, 1e-3, 1e-4, 1e-5\}$. Fig. 16 presents a quantitative evaluation of the influence of varying lr values on classification accuracy. A clear overall trend is that as lr increases, OA first increases and then decreases. The best classification accuracy occurs when lr = $1e-2$. Thus, we set lr to $1e-2$ uniformly across all experiments.

B. Ablation Experiments

Ablation studies were conducted to validate the effectiveness of key learnable modules. Our analysis covers different metrics, GAM, CMA, and CMI modules. The baseline

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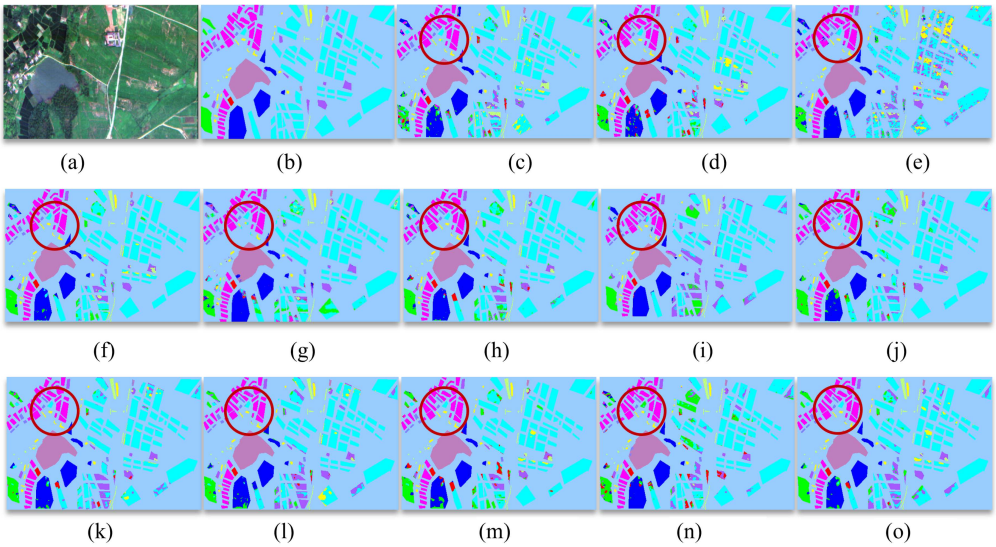


Fig. 13. Visual representations of various classification algorithms on the Tea Farm dataset. (a) False-color image. (b) Ground truth. (c) 3DDL. (d) SSFTT. (e) S3Net. (f) GAHT. (g) GTFN. (h) DCFSL. (i) HFSL. (j) Gia-CFSL. (k) CMFSL. (l) RCA-PDA. (m) STBDIP. (n) ADAFSL. (o) TeFL.

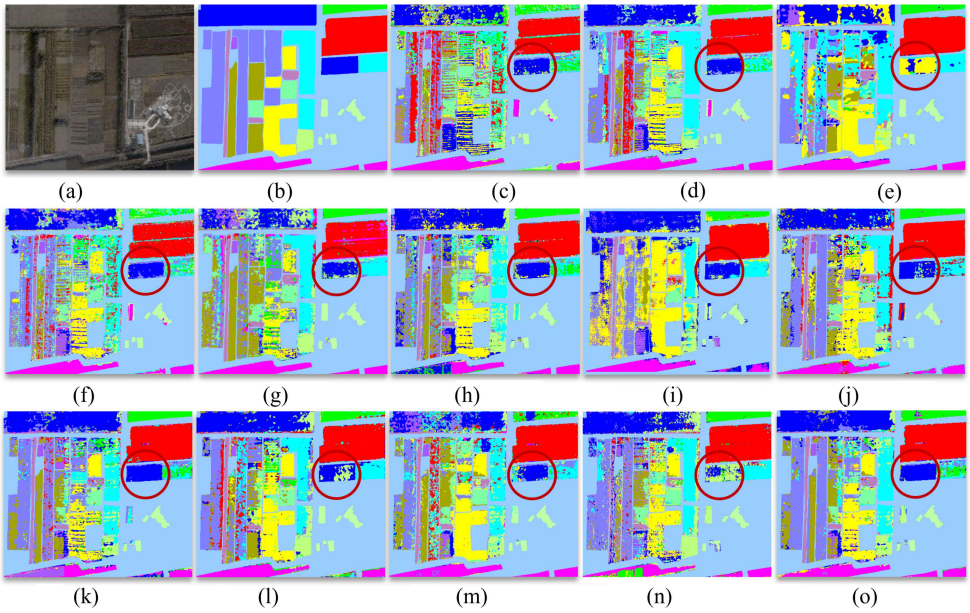


Fig. 14. Visual representations of various algorithms on the Xiong An dataset. (a) False-color image. (b) Ground truth. (c) 3DDL. (d) SSFTT. (e) S3Net. (f) GAHT. (g) GTFN. (h) DCFSL. (i) HFSL. (j) Gia-CFSL. (k) CMFSL. (l) RCA-PDA. (m) STBDIP. (n) ADAFSL. (o) TeFL.

TABLE X

EVALUATING THE CONTRIBUTION OF DIFFERENT LEARNABLE MODULES ON DIVERSE DATASETS

CS	ED	GAM	FSL	CMA	CMI	GFF-A		GFF-B		Tea Farm		Xiong An	
						OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa
✓	×	×	✓	×	×	74.33±2.05	71.77±2.25	74.35±4.04	68.86±4.70	86.08±3.83	82.73±4.95	68.63±5.77	64.53±6.35
×	✓	×	✓	×	×	78.62±3.19	76.58±3.51	81.03±2.32	76.99±1.56	91.00±2.18	87.98±2.98	73.79±2.80	68.60±3.11
×	✓	✓	✓	×	×	79.43±2.06	77.40±2.26	82.40±2.37	78.31±2.74	92.76±2.72	89.67±3.67	74.09±2.75	69.67±3.02
×	✓	✓	✓	✓	×	79.92±2.27	77.92±2.50	83.42±1.96	79.52±2.30	93.26±2.15	90.34±2.92	76.63±2.97	73.42±3.27
×	✓	✓	✓	✓	✓	**80.67±1.85**	**78.73±2.04**	**83.76±1.86**	**79.93±2.20**	**93.29±1.67**	**90.39±2.29**	**76.96±2.75**	**73.76±3.03**

employs standard 3-D convolutions without any attention mechanisms or learnable units. Table X quantifies the contributions of the various modules. Compared to CS, ED demonstrates superior performance in hyperspectral high-dimensional metric contexts. GAM can obtain discriminative visual features through inexpensive operations, boosting OA by 0.81% (GFF-A), 1.37% (GFF-B), 1.76% (Tea Farm), and 1.3% (Xiong An). The gradual integration of the CMA and CMI modules further enhances accuracy, validating their practicality and effectiveness. Moreover, we visualize the

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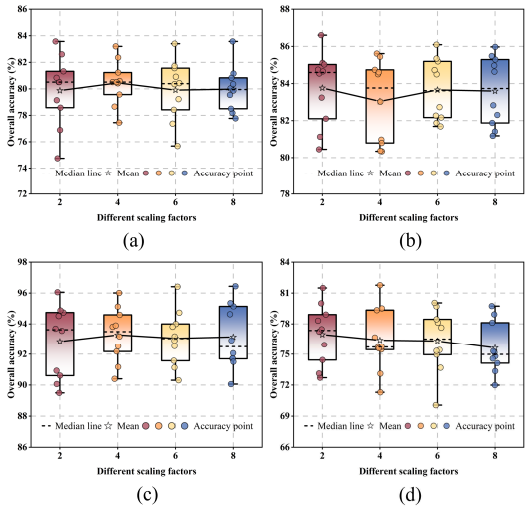


Fig. 15. Quantitative assessment of various scaling factors r on diverse datasets. (a) GFF-A. (b) GFF-B. (c) Tea Farm. (d) Xiong An.

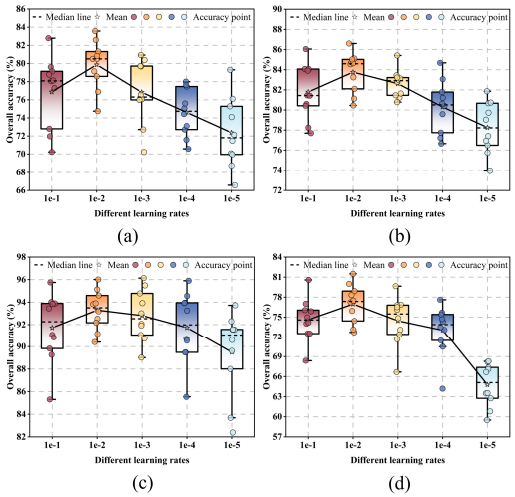


Fig. 16. Quantitative analysis of various learning rates lr on diverse datasets. (a) GFF-A. (b) GFF-B. (c) Tea Farm. (d) Xiong An.

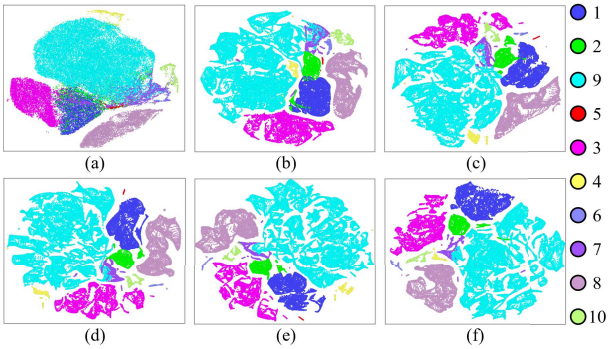


Fig. 17. Embedding feature visualization on the Tea Farm dataset via t-SNE, where different colors represent distinct classes. (a) Original distribution. (b) Baseline model (CS). (c) Baseline model (ED). (d) TeDFL (ED+GAM). (e) TeDFL (ED+GAM+CMA). (f) TeDFL (ED+GAM+CMA+CMI).

high-dimensional output features under different module combinations using t-SNE, as shown in Fig. 17. We find that our modules can extract discriminative information by enhancing intraclass feature aggregation and interclass feature separation (e.g., Class 1). Synthesizing both quantitative and qualitative results, we achieve optimal classification performance when all proposed modules are utilized.

TABLE XI
PERFORMANCE COMPARISON OF DIVERSE PRETRAINED CLIP MODELS FOR TARGET DATASETS

Datasets	Metrics	Different pretrained CLIP models				
		RN50	RN101	ViT-B/32	ViT-B/16	ViT-L/14
GFF-A	OA	79.84±1.81	**80.67±1.86**	79.00±3.37	79.94±1.38	79.77±2.41
	AA	79.92±1.80	**80.74±1.85**	79.09±3.35	80.02±1.38	79.85±2.40
	Kappa	77.83±2.00	**78.73±2.04**	76.90±3.71	77.93±1.52	77.75±2.65
GFF-B	OA	**83.76±1.86**	83.45±1.84	82.93±1.24	83.39±1.99	83.11±2.03
	AA	86.23±1.80	**86.72±1.47**	85.54±1.56	86.18±1.77	85.62±1.41
	Kappa	**79.93±2.20**	79.58±2.15	78.89±1.42	79.47±2.34	79.13±2.42
Tea Farm	OA	92.73±2.33	92.39±2.87	92.84±2.45	93.15±2.02	**93.29±1.67**
	AA	93.18±1.70	93.30±2.48	93.38±2.13	93.81±1.91	**94.35±1.59**
	Kappa	89.62±3.17	89.18±3.91	89.79±3.33	90.19±2.75	**90.39±2.29**
Xiong An	OA	75.27±2.66	76.34±2.90	76.35±3.08	76.23±2.84	**76.96±2.75**
	AA	79.42±1.52	79.99±1.45	80.24±1.83	**80.91±1.80**	80.67±1.49
	Kappa	71.88±2.86	73.06±3.22	73.09±3.36	72.99±3.16	**73.76±3.03**

C. Impact of Different Pretrained CLIP Models on Classification Performance

The CLIP model has two primary structural variants: ResNet-based and ViT-based. Though they share an identical text encoder, parametric and representational differences of the encoder lead to its varying adaptabilities to task requirements and computational constraints. Thus, we conduct experiments on each dataset using the following five pretrained models: RN50, RN101, VIT-B/32, VIT-B/16, and VIT-L/14. Table XI presents the classification results of these pretrained models on each tree species dataset. On the GFF-A and GFF-B datasets, ResNet-based architectures generally outperform those based on ViT. In contrast, ViT-based CLIP models exhibit superior performance on the Tea Farm and Xiong An datasets, with the ViT-L/14 pretrained model achieving the best results.

D. Analysis of the Number of Target Labeled Samples

The amount of labeled data used in the training process significantly impacts classification accuracy. Therefore, we randomly select L labeled samples from each class in the target datasets to thoroughly evaluate model performance. The values of L considered range from the set $\{1, 2, 3, 4, 5\}$. Fig. 18 displays a comparison of classification performance among different models under varying quantities of target labeled samples. As anticipated, the OA of all models increases as the value of L rises. Among the various comparative algorithms, CMFSL, STBDIP, and DCFSL exhibit strong competitiveness with different quantities of target labeled samples. Notably, the proposed TeDFL method demonstrates stronger classification strength than other comparative methods in all scenarios, even with fewer labeled samples.

E. Influence of Diverse Source-Domain Datasets on Classification Accuracy

To investigate the impact of different source-domain datasets on the classification model, we conducted additional experiments on the Botswana (14 classes, 1476 × 256 pixels), Dioni

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TABLE XII
COMPUTATIONAL TIME FOR VARIOUS MODELS ON DIVERSE DATASETS (WITH FIVE LABELED SAMPLES PER CLASS)

Datasets	Metrics	Non-FSL-based methods					FSL-based methods							
		3DDL	SSFTT	S3Net	GAHT	GTFN	DCFSL	HFSL	Gia-CFSL	CMFSL	RCA-PDA	STBDIP	ADAFSL	TeDFL
GFF-A	Training time (s)	49.78	82.16	75.29	134.60	161.93	3085.31	617.19	2752.47	1572.92	1877.96	2401.61	28540.32	1677.48
	Testing time (s)	0.07	0.10	0.67	0.11	0.88	3.05	0.98	0.15	0.21	0.22	0.21	0.18	0.14
GFF-B	Training time (s)	142.26	78.09	18.51	64.04	2994.76	4842.51	983.71	3826.35	2625.01	2965.68	5241.74	36043.83	2331.19
	Testing time (s)	15.80	17.72	75.86	17.78	334.93	39.61	123.75	57.52	17.64	63.70	32.50	53.94	13.84
Tea	Training time (s)	12.63	16.05	12.32	18.86	591.02	1252.00	320.29	3421.57	1401.18	2112.62	1792.04	22696.99	1133.83
	Testing time (s)	3.01	1.80	11.97	2.95	60.73	23.24	19.25	7.79	5.76	10.89	11.24	23.02	2.97
XA	Training time (s)	17.35	31.49	23.02	20.71	1791.66	4690.26	1709.35	5885.72	2534.75	4483.16	9721.16	31387.97	2787.10
	Testing time (s)	8.51	12.50	107.96	15.17	182.74	207.72	56.41	65.30	64.38	83.45	73.94	48.24	18.34

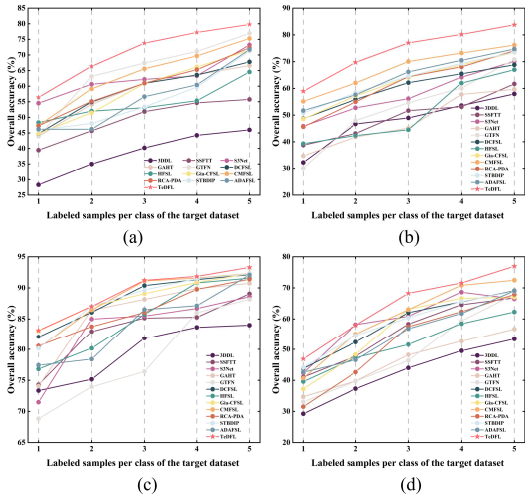


Fig. 18. Impact of labeled sample size on OA for different target datasets. (a) GFF-A. (b) GFF-B. (c) Tea Farm. (d) Xiong An.

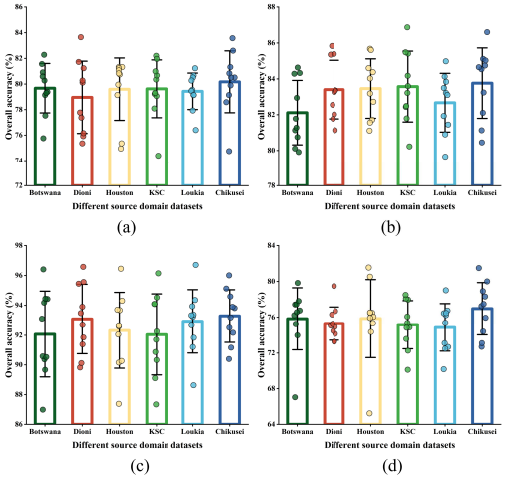


Fig. 19. Classification performance of various source datasets on novel datasets. (a) GFF-A. (b) GFF-B. (c) Tea Farm. (d) Xiong An. The number of dots per group indicates the cross-experiment count.

(12 classes, 250 × 1376 pixels), Houston (15 classes, 349 × 1905 pixels), KSC (13 classes, 512 × 614 pixels), and Loukia (12 classes, 249 × 945 pixels) datasets. Compared to these datasets, Chikusei has more classes and a larger sample size. To ensure cross-experimental consistency, we restricted labeled target samples to five per class. Fig. 19 demonstrates how various source datasets influence the OA of target datasets. Notably, when the Chikusei dataset is used as the source dataset, the TeDFL model delivers better

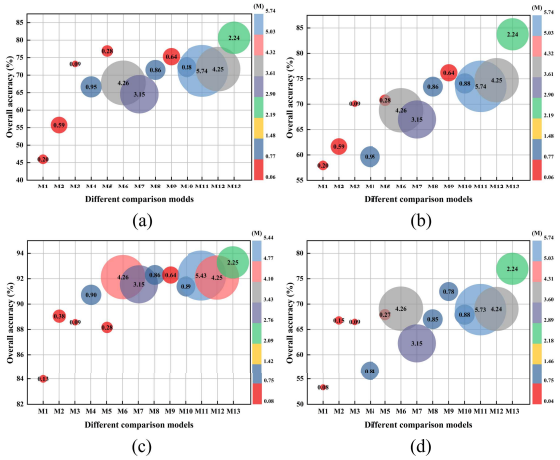


Fig. 20. Classification performance and parameters of different models on datasets. (a) GFF-A. (b) GFF-B. (c) Tea Farm. (d) Xiong An. Larger circles mean more parameters, higher circles mean higher accuracy, and vice versa.

overall classification performance. Consequently, the variety and abundance of source domain samples are critical factors that significantly impact model capability, especially for tree species classification tasks with limited data.

F. Computational Complexity Evaluation

Fig. 20 compares training parameter counts across models via circle size. Models are labeled M0-M13, where M13 represents TeDFL. Fig. 20 illustrates that the non-FSL-based methods like 3DDL and S3Net require fewer parameters, yet exhibit limited performance, while FSL-based methods such as DCFSL, STBDIP, and ADAFSL demand more parameters due to their deep residual or multiscale modules. Our TeDFL method balances parameter efficiency and performance. We have conducted a comprehensive collation of the training and inference times across different models, as presented in Table XII. The training and inference times of TeDFL are shorter than those of most FSL-based comparative methods, and its inference speed is highly competitive when compared to non-FSL-based methods. Overall, the TeDFL method excels in both classification accuracy and computational efficiency.

V. CONCLUSION

In this study, we propose a TeDFL framework that integrates visual and textual features for few-shot tree species classification. First, we construct the GANet visual encoder to extract rich discriminative visual features with few training parameters. Subsequently, the CLIP model is used to incorporate

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linguistic prior information to provide a more comprehensive understanding of various tree species. Additionally, the CMA and CMI modules are specifically designed to address inherent discrepancies across different modalities. The core function of the CMA module is to align visual and textual features through contrastive learning, while the CMI module aims to improve the quality of class prototypes through multimodal prototype classification constraints. Extensive experiments and analysis on four forest datasets validate TeDFL's superior classification performance under label-scarce conditions. In the next stage, we will explore more general and effective forms of text prompts and combine them with additional data (e.g., LiDAR) to further investigate the adaptability of VLMs in large-scale tree species mapping and their generalization across different remote sensing tasks.

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